# DECENTRALISED MULTI-AGENT SEARCH AND RESCUE UNDER LIMITED COMMUNICATION

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## ABSTRACT

A person is trapped at an unknown location inside a hazardous building, and a team of heterogeneous autonomous agents must locate them and bring them to an exit. Search drones sense well but cannot perform the rescue; ground rescue robots can perform the rescue but sense poorly, so information must travel from the agents that locate the victim to the agents that can reach them. Communication, however, is a limited resource. We model the building as a graph with obstacles, hazards, an exit and one hidden victim; agents keep Bayesian beliefs built from noisy local scans and allocate tasks under a team communication budget $B$. A centralised baseline solves the task-allocation CSP exactly with the Hungarian method on pooled information, and a decentralised method solves the corresponding DCOP with the Maximum Gain Message (MGM) algorithm over a range-limited communication graph. Over 25,200 simulated episodes we find that (i) the first $\approx$2 messages per agent—the drone-to-robot handoff—recover more than half of the performance lost without communication (success $0.74 \rightarrow 0.89$); (ii) with unrestricted budget the decentralised team uses 44% of the centralised message volume and is only 9% slower; (iii) communication is worth more when sensors are noisy, but small budgets also make robots chase false alarms; and (iv) coordination (MGM) messages only pay off once the budget is large enough—below $\approx$40 messages they crowd out observation sharing and an uncoordinated variant does better.

***Keywords*** Multi-agent systems Search and rescue Decentralised coordination Task allocation Constraint satisfaction Distributed constraint optimisation Limited communication

# 1 Introduction

Search-and-rescue operations in hazardous buildings are difficult because responders often do not know where a trapped person is, or which routes remain safe. Autonomous robots can support such operations by exploring, sensing and mapping, and multi-robot teams have been studied extensively for exactly this setting [1]. Teams that mix robots with different capabilities have also been studied for urban search and rescue, where the question of which robot should take which task becomes central [3].

This project considers a team of heterogeneous agents inside a simulated hazardous building: *search drones*, which move quickly and sense over a wide area but cannot rescue anyone, and *ground rescue robots*, which move only along accessible ground routes and sense poorly but can carry the victim to an exit. The split is deliberate: the agents that find the victim are never the agents that can

rescue the victim, so a report has to travel from one group to the other. Three properties make this a multi-agent problem rather than a path-planning one: no agent observes the whole environment; capabilities are complementary, so tasks must be allocated to agents able to perform them [4]; and communication is limited, so agents must decide not only where to go but also what is worth saying. Such coordination is naturally modelled as a distributed constraint optimisation problem (DCOP) [2].

**Research question.** *How does reducing the communication budget affect decentralised coordination in multi-agent search and rescue when agents have incomplete and noisy information?* Secondary questions are whether communication becomes more or less valuable as sensors become less reliable, how far decentralised coordination stays from a centralised coordinator with pooled information, and how much redundant exploration [6] appears when agents cannot talk. This report completes the project review with the methodology (Section 4), the results (Section 5) and the individual contributions (Section 6).

# 2 Preliminaries

**Agents and heterogeneity.** $\mathscr{A} = \mathscr{D} \cup \mathscr{R}$, $\mathscr{D} \cap \mathscr{R} = \emptyset$, where $\mathscr{D}$ are search drones and $\mathscr{R}$ ground robots. Drones have a larger sensing radius and higher reliability; only robots can rescue. **Environment.** The building is a graph $G = (V, E)$ generated as a 2D grid whose cells are free, obstacles or hazardous, with one exit $v_{\text{exit}}$ and one victim location $v^\star$ unknown to the agents; time is discrete. **Incomplete information.** A *local observation* is what an agent's sensors return at its location; it is *noisy*: a scan may miss the victim or report one where there is none. **Tasks.** $\mathscr{T}$ is the set of candidate tasks (searching a region, carrying out the rescue) with rescue tasks $\mathscr{T}_{\text{res}} \subset \mathscr{T}$; *task allocation* assigns tasks to capable agents [4]. **Communication.** Agents exchange short messages; the total they may send in a run is limited by a budget $B$. **CSP and DCOP.** A CSP is a triple of variables, domains and constraints; a DCOP distributes the variables among agents and replaces constraints by cost functions that agents minimise by exchanging messages with neighbours [2].

# 3 Problem Formulation

Let $p_i^t \in V$ be the location of agent $i$ at time $t$. **Decision variables:** $x_{ij} = 1$ if task $j \in \mathscr{T}$ is assigned to agent $i \in \mathscr{A}$ and 0 otherwise. **Assignment constraints:**

$$\textstyle\sum_{i \in \mathscr{A}} x_{ij} \le 1 \;\; \forall j \in \mathscr{T}, \tag{1}$$

$$x_{ij} \le F_{ij} \;\; \forall i, j, \qquad F_{ij} = 0 \;\; \forall i \in \mathscr{D},\, j \in \mathscr{T}_{\text{res}}, \tag{2}$$

where $F_{ij} \in \{0, 1\}$ is agent $i$'s capability for task $j$. **Observation model.** A scan by agent $i$ returns, for every $v$ within radius $\rho_i$, a reading $z_v \in \{0, 1\}$ with

$$\Pr(z_v = 1 \mid v = v^\star) = p_i^{\text{det}}, \qquad \Pr(z_v = 1 \mid v \neq v^\star) = p_i^{\text{fa}}, \tag{3}$$

with $p_i^{\text{det}} > p_{i'}^{\text{det}}$ and $\rho_i > \rho_{i'}$ for $i \in \mathscr{D}, i' \in \mathscr{R}$. **Communication constraint.** $C_{\text{comm}} = \sum_{t=1}^{T} \sum_{i \in \mathscr{A}} c(m_i^t) \le B$. **Objective.** With completion time $T_{\text{rescue}}$, movement cost $C_{\text{move}}$ and success $S \in \{0, 1\}$ ($T_{\text{rescue}}$ is set to the step limit when $S = 0$),

$$\min_{\{x_{ij}\}} \; J = \lambda_t T_{\text{rescue}} + \lambda_c C_{\text{comm}} + \lambda_m C_{\text{move}} + \lambda_f (1 - S) \quad \text{s.t. (1)–(2) and } C_{\text{comm}} \le B. \tag{4}$$

**Decentralised form.** The DCOP $\langle \mathscr{A}, X, D, F, \alpha \rangle$ has variables $X = \{x_{ij}\}$, domains $\{0, 1\}$, cost functions $F$ derived from (4) and $\alpha(x_{ij}) = i$; each agent chooses its own variables from local observations and received messages. The centralised problem is the same optimisation with all variables controlled by one coordinator and $C_{\text{comm}}$ not charged.

# 4 Methodology

Everything below is implemented in Python (NumPy/SciPy) in the accompanying repository; module names are given in `typewriter`. Both approaches run on the *same* simulator and the *same* randomly generated building for a given seed, and share the cost functions, so the only differences are who holds which information and how the allocation is solved.

## 4.1 Simulator (`sar/env.py`, `sar/agent.py`)

**Building.** An $n \times n$ grid ($n = 20$ by default) receives a few internal walls with door gaps (room structure), random debris obstacles (density 0.08) and hazard cells (density 0.06). The exit lies on the left wall; the victim is placed uniformly among robot-reachable cells at least $0.6n$ steps from the exit. Drones fly over hazards but not through walls; robots may not enter hazards. All-pairs shortest-path distances for both movement models are pre-computed with BFS, and agents move along shortest paths (drones 2 cells/step, robots 1 cell/step). The floor plan is known (from blueprints); the victim location is not. **Sensing and belief.** Drones scan a $5 \times 5$ window ($\rho = 2$), robots a $3 \times 3$ window ($\rho = 1$), with the reliability presets of Table 1. Each agent keeps a posterior

$b_i(v) = \Pr(v^\star = v \mid \text{evidence})$, initially uniform over robot-reachable cells. Since there is a single victim, a reading $z_v$ from a sensor with parameters $(p^{\text{det}}, p^{\text{fa}})$ multiplies $b(v)$ by the likelihood ratio

$$L(z_v) = \left(\frac{p^{\text{det}}}{p^{\text{fa}}}\right)^{z_v} \left(\frac{1-p^{\text{det}}}{1-p^{\text{fa}}}\right)^{1-z_v} \tag{5}$$

followed by renormalisation (all other cells share the same denominator). Updates are kept in log space. A robot that stands on a cell physically checks it: if the victim is there it picks them up, otherwise the cell's posterior is set to zero (certain absence). An agent *claims* the victim at $\hat{v} = \arg\max_v b_i(v)$ when $b_i(\hat{v}) \geq \tau = 0.7$.

## 4.2 Communication model and messaging rule (`sar/sim.py`)

A message is a broadcast received by every agent within Euclidean range $r_c = 9$ cells; each costs one unit ($c(m) = 1$). The team budget is split equally, each agent holding a quota $\lfloor B/|\mathscr{A}| \rfloor$, so $C_{\text{comm}} \leq B$ holds by construction and needs no global counter. Neighbour discovery (who is in range) is free. As promised in the review, *what* to send is fixed to one simple rule so that the budget remains the variable under study. Each agent buffers its unshared scans; an `OBS` message carries the whole buffer (so receivers apply exactly the update (5) with the sender's sensor parameters), the sender's current task, its cleared cells and its current claim. Messages are sent in priority order: (1) *urgent*: a new or retracted claim, a robot's report that a claimed cell was empty, or the "victim picked up" notice; (2) *delivery*: a drone holding its own claim that has not yet reached a robot flies towards the last known robot position and broadcasts once a robot is in range; (3) *relay*: a newly heard claim is re-broadcast once; (4) *periodic*: every 5 steps the observation buffer is shared; (5) DCOP messages (Section 4.5). Classes (4)–(5) are only sent while an agent has more than a reserve of 2 messages left, so the reserve is kept for victim reports. **Conflicting reports.** Each agent keeps the latest claim per sender and its own claim; it acts on the one with the highest confidence (ties: most recent), and discards claims on cells it knows to be empty.

## 4.3 Task set and cost functions (`sar/allocation.py`)

The map is split into $K = 16$ regions ($4 \times 4$). The task set is $\mathscr{T} = \{\text{SEARCH}_k\}_{k=1}^{K} \cup \{\text{RESCUE}\} \cup \{\text{IDLE}\}$, where IDLE plays the role of "unassigned" and is exempt from (1). For agent $i$ with belief $b_i$, speed $s_i$ and distance $d_i(\cdot)$ in its own movement graph, the unary costs instantiate the time and failure terms of (4):

$$u_i(\text{SEARCH}_k) = \frac{d_i(k)}{s_i} - w_{\text{info}}\, q_i \sum_{v \in \text{region } k} b_i(v), \quad u_i(\text{RESCUE}) = \frac{d_i(\hat{v})}{s_i} - w_{\text{res}}\, b_i(\hat{v}), \quad u_i(\text{IDLE}) = 0, \tag{6}$$

where $q_i = p_i^{\text{det}}(2\rho_i{+}1)^2/25$ is the scanning efficiency and RESCUE exists only when the agent holds a claim $\hat{v}$. $u_i(\text{RESCUE}) = +\infty$ for drones encodes (2). The pairwise cost $w(x_i, x_j) = w_{\text{dup}}\,[x_i = x_j \neq \text{IDLE}]$ is the soft form of (1). Given its task, an agent moves to the cell of its region maximising (posterior mass in its sensing window)$/(1 + 0.1\, d_i)$; a rescuer moves to $\hat{v}$ and, once carrying, to the exit; IDLE agents hold position (this is what drones do once the belief has collapsed, saving movement).

## 4.4 Centralised baseline: exact assignment

A coordinator receives every scan every step (counted as one uplink message per agent per step, plus one downlink per changed assignment, but never limited), maintains one pooled belief, and every 5 steps (or when a claim appears) solves $\min \sum_i u_i(x_i)$ subject to (1)–(2) *exactly*: rows are agents, columns the non-idle tasks plus one private IDLE column per agent, infeasible pairs get cost $10^6$, and the rectangular assignment is solved by the Hungarian method [8] (`scipy.optimize.linear_sum_assignment`). Because constraint (1) is hard, it is never violated (verified in `tests/test_model.py`).

## 4.5 Decentralised DCOP coordination: MGM

The review left the DCOP solver open. We chose MGM [7] over ADOPT [5]: ADOPT is complete but needs a DFS tree and a number of messages exponential in the worst case, which cannot be stopped at an arbitrary budget, whereas MGM is an anytime local search whose cost is linear in rounds, whose solution quality never decreases between rounds, and which degrades gracefully when messages are dropped—the regime we study. Every 5 steps each agent runs the rounds of Algorithm 1 using *its own* belief; tasks of neighbours are remembered in an *agent view* that expires after 15 steps. Between rounds, an agent hit by a new claim re-decides locally. With $B = 0$ no `VALUE`/`GAIN` message is ever sent and the algorithm reduces to independent greedy choices—the no-communication ablation. We also run a **no-MGM ablation** that sends only `OBS` messages (tasks are learned from their piggyback) and lets every agent take its best response, isolating the value of coordination messages from the value of shared observations.

**Algorithm 1** One MGM allocation epoch at agent $i$ (decentralised; all agents run it in parallel)

**input:** belief $b_i$, claim $\hat{v}$ (if any), agent view $\mathcal{V}_i$, quota left $Q_i$
compute unary costs $u_i(\cdot)$ on own domain; if current $x_i$ infeasible: $x_i \leftarrow \arg\min_x [u_i(x) + \sum_{j \in \mathcal{V}_i} w(x, x_j)]$
**for** round $= 1..3$ **do**
  **if** $x_i$ changed or announcement is stale **and** $Q_i >$ reserve: broadcast $\texttt{VALUE}(x_i)$ (cost 1)
  on receiving $\texttt{VALUE}(x_j)$: $\mathcal{V}_i[j] \leftarrow (x_j, t)$
  $x_i^* \leftarrow \arg\min_x \left[u_i(x) + \sum_{j \in \mathcal{V}_i,\, \text{fresh}} w(x, x_j)\right]$; $g_i \leftarrow \text{cost}(x_i) - \text{cost}(x_i^*)$
  **if** $g_i > 0$ **and** $Q_i >$ reserve: broadcast $\texttt{GAIN}(g_i)$ (cost 1)
  **if** $g_i > 0$ and $(g_i, -i) > (g_j, -j)$ for every gain $g_j$ heard this round: $x_i \leftarrow x_i^*$
  **if** no agent changed: **break**

**Table 1:** Default parameters. Reliability is $(p^{\text{det}}, p^{\text{fa}})$ per scanned cell.

| | | | |
|---|---|---|---|
| Grid / regions | $20 \times 20$ / $4 \times 4$ | Team | 3 drones + 2 robots |
| Speed (drone / robot) | 2 / 1 cells per step | Sensing radius (drone / robot) | 2 / 1 |
| High reliability | drone (0.95, 0.01), robot (0.70, 0.02) | Comm. range $r_c$ | 9 cells |
| Medium (default) | drone (0.80, 0.03), robot (0.55, 0.04) | Claim threshold $\tau$ | 0.7 |
| Low reliability | drone (0.60, 0.06), robot (0.45, 0.07) | Replan period / MGM rounds | 5 steps / 3 |
| $w_{\text{info}}, w_{\text{res}}, w_{\text{dup}}$ | 400, 1000, 1000 | Share period / reserve | 5 steps / 2 msgs |

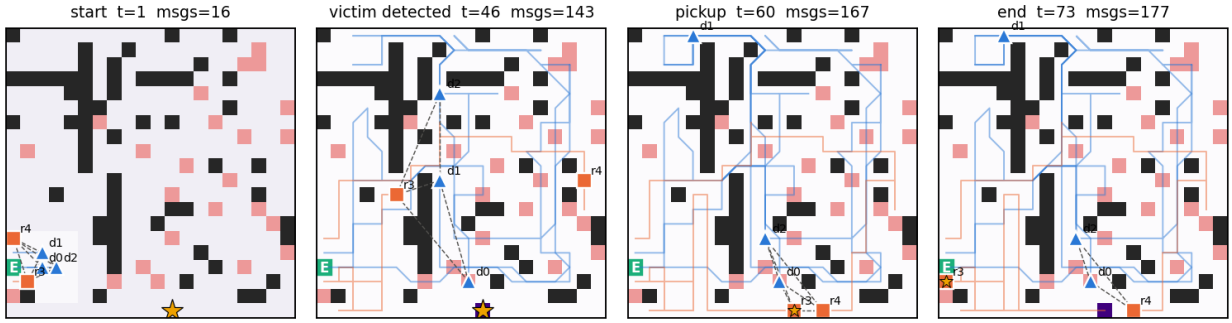



**Figure 1:** A decentralised episode (seed 3, $B = \infty$) from the simulator: obstacles (black), hazards (red), exit (E), drones $d_j$ (blue triangles), robots $r_j$ (orange squares), victim (star), communication links within range (dashed), agent trajectories, and drone $d_0$'s posterior (purple). Drones sweep the building, $d_0$ claims the victim at $t = 46$, robot $r_3$ is told within the same step and carries the victim out at $t = 73$.

### 4.6 Experimental setup (`experiments/run_experiments.py`)

Default parameters are listed in Table 1. Each configuration is run on the same 300 seeds (matched building instances) for every method and budget; we report means with 95% confidence intervals. The objective weights are $\lambda_t = 1$, $\lambda_c = 0.1$, $\lambda_m = 0.05$, $\lambda_f = 300$ and the step limit (survival window) is 150. Experiments: **(E1)** budget $B \in \{0, 10, 20, 40, 80, 160, 320, \infty\}$ for MGM and the no-MGM ablation; **(E2)** E1 at three sensor-reliability levels; **(E3)** team size $(|\mathscr{D}|, |\mathscr{R}|) \in \{(2,1), (3,2), (5,3)\}$ and **(E4)** building complexity (simple $15^2$, medium $20^2$, complex $28^2$ with increasing obstacle/hazard density and walls), both at $B \in \{0, 20, 80, \infty\}$ (none/low/moderate/unrestricted); **(E5)** communication range $r_c \in \{4, 6, 9, 14, \infty\}$ at $B \in \{40, \infty\}$. In total 25,200 episodes; the whole study runs in about four minutes on a 10-core laptop. Besides the four metrics of the review we log the handoff time (first step a robot holds a correct claim), the number of steps in which (1) is violated, robot trips to false claims, and *redundancy*: the fraction of scanned cells scanned by two or more agents.

## 5 Results

### 5.1 E1: performance against the communication budget

Figure 2 and Table 2 answer the main question. Performance does **not** degrade gradually: it drops sharply only at the very bottom of the budget range. Without communication the team succeeds in 74% of runs, robots learn the victim location only after 71 steps on average (they must find the victim with their own poor sensors) and two agents share a task in 77 of the steps. Allowing just $B = 10$ messages—two per agent, which the reserve rule spends on the claim and its relay—cuts the handoff time to 37 steps and

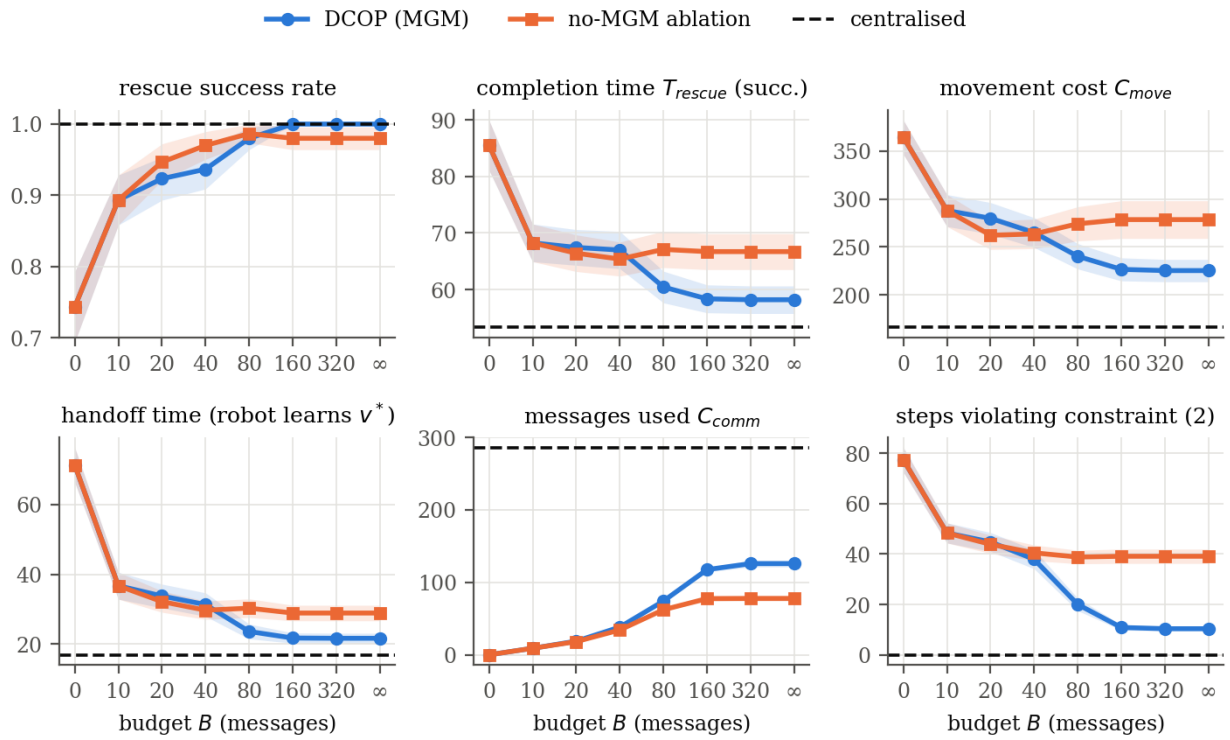



**Figure 2:** E1: metrics against the team budget $B$ (medium reliability, 300 seeds; bands are 95% CIs). Dashed line: centralised reference. Completion time is averaged over successful runs.

**Table 2:** E1 summary (mean $\pm$ 95% CI over 300 matched seeds). $T$ on successful runs; "(2) viol." = steps in which two agents hold the same non-idle task.

| Method | $B$ | Success | $T_{\text{rescue}}$ | $C_{\text{comm}}$ | $C_{\text{move}}$ | (2) viol. | Redund. |
|---|---|---|---|---|---|---|---|
| Centralised | – | $1.00 \pm 0.00$ | $53.3 \pm 2.0$ | $286.0 \pm 10.1$ | $166 \pm 6$ | 0.0 | 0.52 |
| DCOP (MGM) | 0 | $0.74 \pm 0.05$ | $85.5 \pm 4.4$ | $0.0 \pm 0.0$ | $364 \pm 18$ | 77.5 | 0.81 |
| | 10 | $0.89 \pm 0.03$ | $68.2 \pm 3.3$ | $9.0 \pm 0.2$ | $288 \pm 17$ | 48.4 | 0.78 |
| | 20 | $0.92 \pm 0.03$ | $67.4 \pm 3.2$ | $18.8 \pm 0.2$ | $280 \pm 16$ | 44.8 | 0.78 |
| | 40 | $0.94 \pm 0.03$ | $67.0 \pm 3.3$ | $38.2 \pm 0.2$ | $265 \pm 15$ | 37.9 | 0.71 |
| | 80 | $0.98 \pm 0.02$ | $60.4 \pm 2.8$ | $74.1 \pm 0.5$ | $240 \pm 13$ | 20.0 | 0.68 |
| | $\infty$ | $1.00 \pm 0.00$ | $58.2 \pm 2.4$ | $125.8 \pm 4.6$ | $225 \pm 12$ | 10.4 | 0.66 |
| No-MGM | 0 | $0.74 \pm 0.05$ | $85.5 \pm 4.4$ | $0.0 \pm 0.0$ | $364 \pm 18$ | 77.5 | 0.81 |
| | 10 | $0.89 \pm 0.03$ | $68.2 \pm 3.3$ | $9.0 \pm 0.2$ | $288 \pm 17$ | 48.4 | 0.78 |
| | 20 | $0.95 \pm 0.03$ | $66.4 \pm 3.2$ | $18.2 \pm 0.2$ | $262 \pm 15$ | 43.9 | 0.78 |
| | 40 | $0.97 \pm 0.02$ | $65.4 \pm 3.0$ | $34.2 \pm 0.4$ | $264 \pm 15$ | 40.5 | 0.79 |
| | 80 | $0.99 \pm 0.01$ | $67.1 \pm 3.2$ | $62.1 \pm 1.5$ | $274 \pm 18$ | 38.8 | 0.79 |
| | $\infty$ | $0.98 \pm 0.02$ | $66.7 \pm 3.2$ | $77.8 \pm 3.6$ | $278 \pm 20$ | 39.1 | 0.79 |

raises success to 89%, recovering 58% of the success gap to unrestricted communication. Beyond that, returns diminish: success reaches 98% at $B = 80$ and 100% at $B = 160$, and the budget stops binding at about 125 messages (the curves for 160, 320 and $\infty$ coincide). Communication also removes redundant exploration: the share of cells scanned by several agents falls from 0.81 to 0.66 and movement cost falls by 38%.

**Gap to the centralised reference.** With unrestricted budget the decentralised team is 9% slower ($T = 58.2$ vs. 53.3), moves 36% more and still has some duplicated assignments (10 steps), because information travels only through range-limited broadcasts and allocation is a local optimum. It does so with 44% of the centralised message volume (126 vs. 286), and its objective $J$ (which charges messages) is lower (82.0 vs. 90.2). Figure 6 shows the resulting trade-off curve.

**Coordination vs. information messages (no-MGM ablation).** With a generous budget MGM is clearly useful: it lowers completion time from 68.4 to 58.2 and duplicated assignments from 39 to 10 steps. Under a tight budget the picture reverses: at

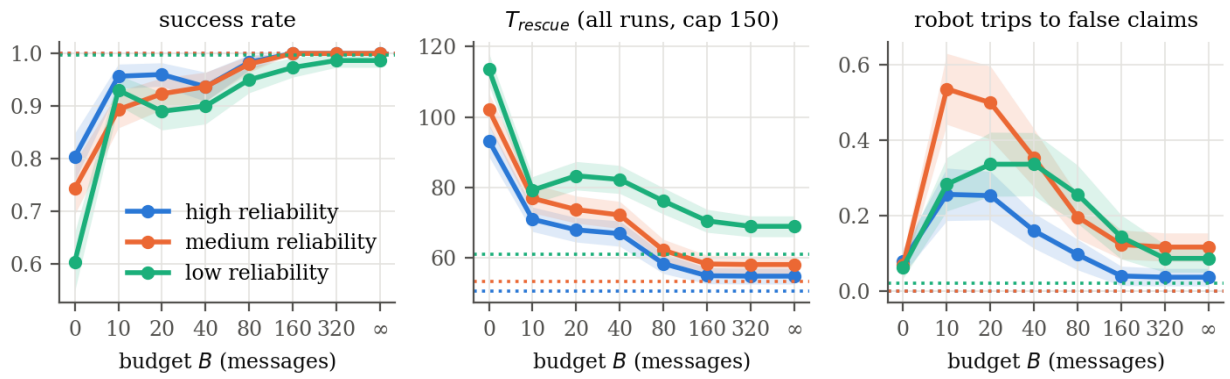



**Figure 3:** E2: budget sweep at three sensor-reliability levels. Dotted lines: centralised reference of the same colour.

**Table 3:** E2: effect of sensor reliability. Success rate and $T_{\text{rescue}}$ (all runs, failures counted at the 150-step limit); last two columns at $B = \infty$.

| | Success | | | | $T_{\text{rescue}}$ | | | | wrong | false |
|---|---|---|---|---|---|---|---|---|---|---|
| Reliab. | Central | $B{=}\infty$ | $B{=}20$ | $B{=}0$ | Central | $B{=}\infty$ | $B{=}20$ | $B{=}0$ | claims | trips |
| high | 1.00 | 1.00 | 0.96 | 0.80 | 50.7 | 54.9 | 68.0 | 93.3 | 0.21 | 0.04 |
| medium | 1.00 | 1.00 | 0.92 | 0.74 | 53.3 | 58.2 | 73.8 | 102.1 | 0.82 | 0.12 |
| low | 1.00 | 0.99 | 0.89 | 0.60 | 61.2 | 69.0 | 83.4 | 113.5 | 0.65 | 0.09 |

$B = 20$–$40$ the uncoordinated variant succeeds more often (0.95/0.97 vs. 0.92/0.94). The message breakdown explains why: at $B = 20$ MGM spends 9.5 of its 18.8 messages on `VALUE`/`GAIN` and only 9.3 on observations, whereas the ablation spends all 18.2 on observations. When messages are scarce, sharing evidence is worth more than negotiating who searches where.

## 5.2 E2: is communication more valuable when sensors are noisy?

Yes, in aggregate (Figure 3, Table 3). The success gap between no communication and unrestricted communication grows from 0.20 (high reliability) to 0.26 (medium) and 0.39 (low), and the time gap to the centralised reference grows from 4.2 to 7.8 steps. With noisy sensors no single agent can confirm the victim quickly, so pooling evidence across agents matters more. The secondary hypothesis of the review—that sharing also spreads false alarms—is confirmed in a specific form (right panel of Figure 3): robot trips to empty claimed cells peak at *small* budgets ($B = 10$–$40$), where a claim is transmitted but most of the evidence that would later refute it is not, and they vanish as $B$ grows. Under low reliability this makes the curve non-monotone: $B = 10$ (only claims are sent) reaches 0.93 success whereas $B = 20$–$40$ reach 0.89–0.90, because periodic sharing of noisy scans at those budgets produces more wrong claims (1.34 per run at $B = 20$ vs. 0.72 at $B = 10$). This is the negative result anticipated in the review: a little noisy information can be worse than none beyond the victim report itself.

## 5.3 E3–E5: is the trend specific to one team or one map?

The shape of the curve—large loss at $B = 0$, most of it recovered by a small budget—holds for every team size and map (Figure 4). **Team size:** the smallest team suffers most without communication (success 0.55 for 2D+1R vs. 0.81 for 5D+3R) because a single robot must find the victim alone; with any budget all teams exceed 0.85. The centralised message volume grows linearly with team size (198→433) while the decentralised volume at $B = \infty$ stays below half of it (81→199). **Complexity:** the gap widens with the building. On the simple map even $B = 0$ succeeds in 94% of runs; on the complex $28 \times 28$ map success falls to 0.42 without communication and the decentralised team stays below the centralised one even with unrestricted budget (0.90 vs. 0.97), because the fixed 9-cell range now covers a smaller fraction of the building and claims must be physically carried to the robots. **Communication range (Figure 5):** with $B = \infty$, widening the range steadily closes the gap to the centralised baseline ($T = 66.1$ at $r_c = 4$ to 55.3 with global broadcast vs. 53.3 centralised); with $B = 40$ range barely matters (71–75 steps), i.e. once the budget is binding it, not connectivity, is the bottleneck.

## 5.4 Discussion and limitations

Answering the research question: in this setting the value of communication is concentrated in a few messages that carry the drone-to-robot handoff; the B=0 ablation shows that complementary agent types are almost useless without it (drones find the victim at step 25 on average but robots only learn it at step 71). Further budget mainly buys less redundant exploration and better

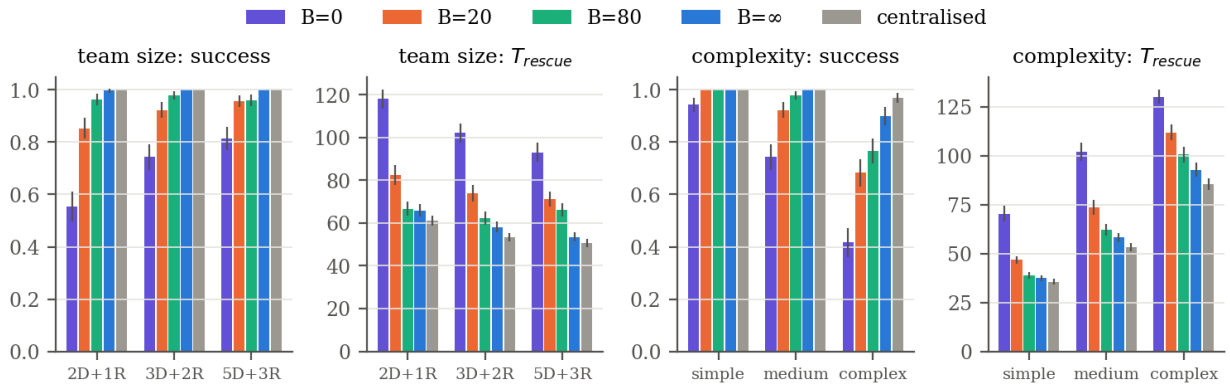



**Figure 4:** E3/E4: success and completion time across team sizes and building complexity at four budgets and for the centralised reference (95% CIs).

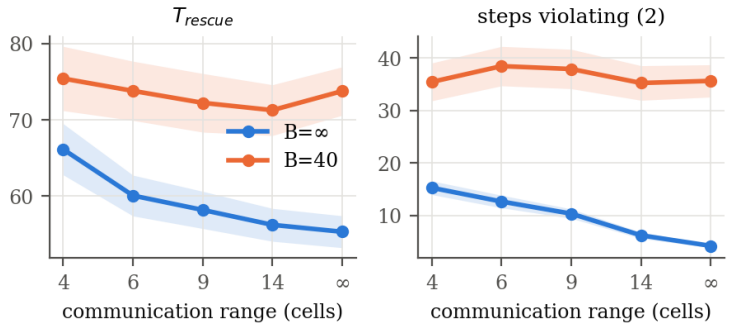



**Figure 5:** E5: completion time and constraint-(2) violations against communication range.

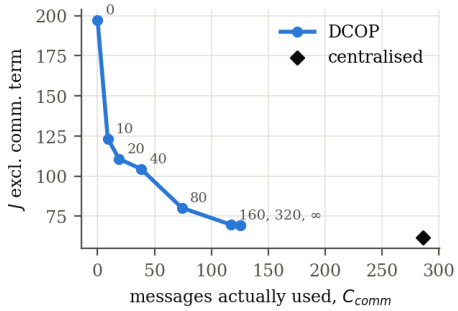



**Figure 6:** Trade-off: messages used vs. the non-communication part of $J$; labels are budgets.

coordination, and the decentralised team then stays within 10% of a centralised coordinator using less than half of its messages. Limitations: the floor plan is known and static; a single victim and abstract, independent sensor noise are assumed; messages are lossless within range; the messaging rule is fixed and its priorities (claims first, observations, then MGM) clearly matter at low budgets, as the no-MGM ablation shows, so a budget-aware rule that decides between information and coordination messages is the natural next step. MGM reaches local optima only, and the numbers are specific to our cost weights; they are simulation results, not claims about real emergency-response systems.

# 6 Contribution

All tasks assigned in the project review were completed; each member also contributed to integration, testing, debugging and writing.

- **Boinapally Dhanvini** — literature review (multi-agent SAR, heterogeneous teams, CSP/DCOP) and the formal model of Section 3; chose MGM over ADOPT and derived the task set and cost functions from (4). *Status: fulfilled.*
- **Bollineni Snehitha Rao** — simulator (`sar/env.py`, `sar/agent.py`): building generation, obstacles, hazards, movement models, the observation model (3), Bayesian belief, and the reliability and complexity presets. *Status: fulfilled.*
- **Jils Shah** — centralised Hungarian baseline and decentralised MGM coordination (`sar/allocation.py`, `sar/sim.py`); communication budget, messaging rule and identical message accounting for both methods; unit tests checking constraints (1)–(2) and the budget. *Status: fulfilled.*
- **Thaker Shivangi Pareshkumar** — experimental design and analysis (`experiments/`): the five experiments with matched seeds, metrics, plots and tables, and the interpretation in Section 5. *Status: fulfilled.*

# 7 Conclusion

We built a simulator of heterogeneous search and rescue under a communication budget and compared an exact centralised task allocation with a decentralised MGM-based DCOP over 25,200 matched episodes. The main finding is that performance falls sharply only when communication is (almost) removed: a couple of messages per agent, used for the drone-to-robot handoff, recover

most of it, and an unrestricted decentralised team comes within 9% of the centralised completion time with 44% of its messages. Communication matters more when sensors are noisy, smaller teams and larger buildings depend on it more, and at small budgets observation sharing should take priority over coordination messages. Code, data, figures and instructions to reproduce every number are in the repository.

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